

# **Orchestrating enterprise infrastructure from a single Cloud Native Control Plane**

Cloud Native Bern Meetup

May 2026



# Facts & Figures

Julius Baer is a leading pure-play Swiss global wealth manager, offering comprehensive services, investment solutions, and wealth planning to private clients and institutions worldwide.

- Founded in Zurich in 1890s
- 7'400 employees in 25 countries
- IT hubs in CH, Spain, Luxembourg and Singapore
  - Classic on-prem infrastructure footprint
  - Mainly Linux & Windows VMs (VMware)
  - Tanzu Kubernetes Platform (TKGI)
- Developer platform on Azure
  - 800 Developers and Application Responsibles
- Production workloads on Azure (NEW)
  - Moving first business applications this year

# Our Journey

# Our automation journey: Chapter 3

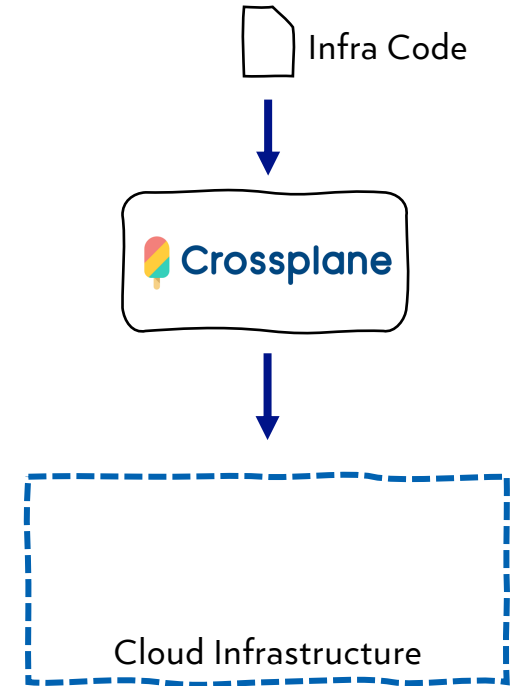
## Why we chose Crossplane to build our control plane

- We evaluated various Infra as Code tools
  - Ansible
  - Azure Bicep
  - Terraform
  - Crossplane

### Key requirements

- Offer a **well-known format** (inspired by the K8S API) to our users
- Cloud **provider agnostic** solution for multi-cloud capability
- **Declarative** over imperative approach
- Continuous (online) state management (unlike Terraform)
- Marketplace of existing providers maintained by a **large community**

We chose **Crossplane** because it ticked all the boxes for us



# Our automation journey: Chapter 4

## Decentralized control plane

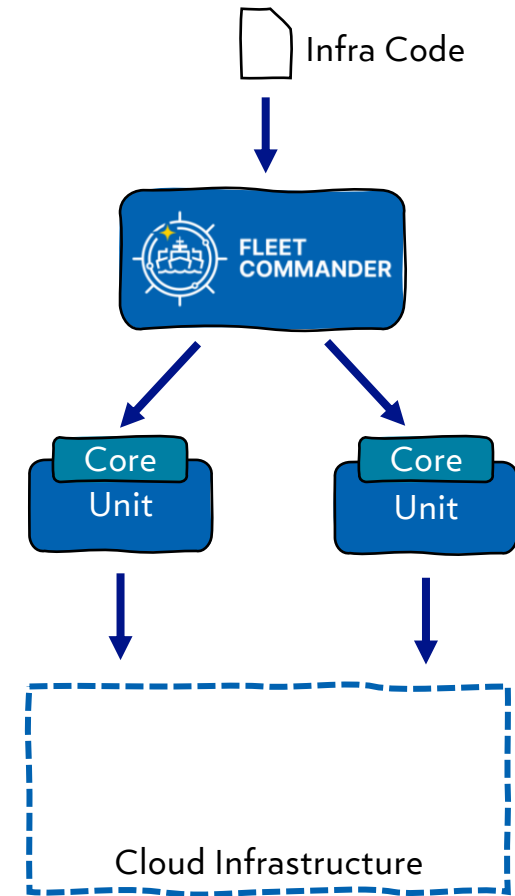
### Apply our learnings to control plane architecture

- ✓ Avoid the bottleneck of a central team
- ✓ Maintain a decentralized automation approach
- ✓ Eliminate API sprawl and bespoke APIs
- ✓ Standardized integration with technology domains

Crossplane with decentralized architecture = Fleet Commander

Commander + Core → **centrally managed**

Units → **decentralized**



# How?

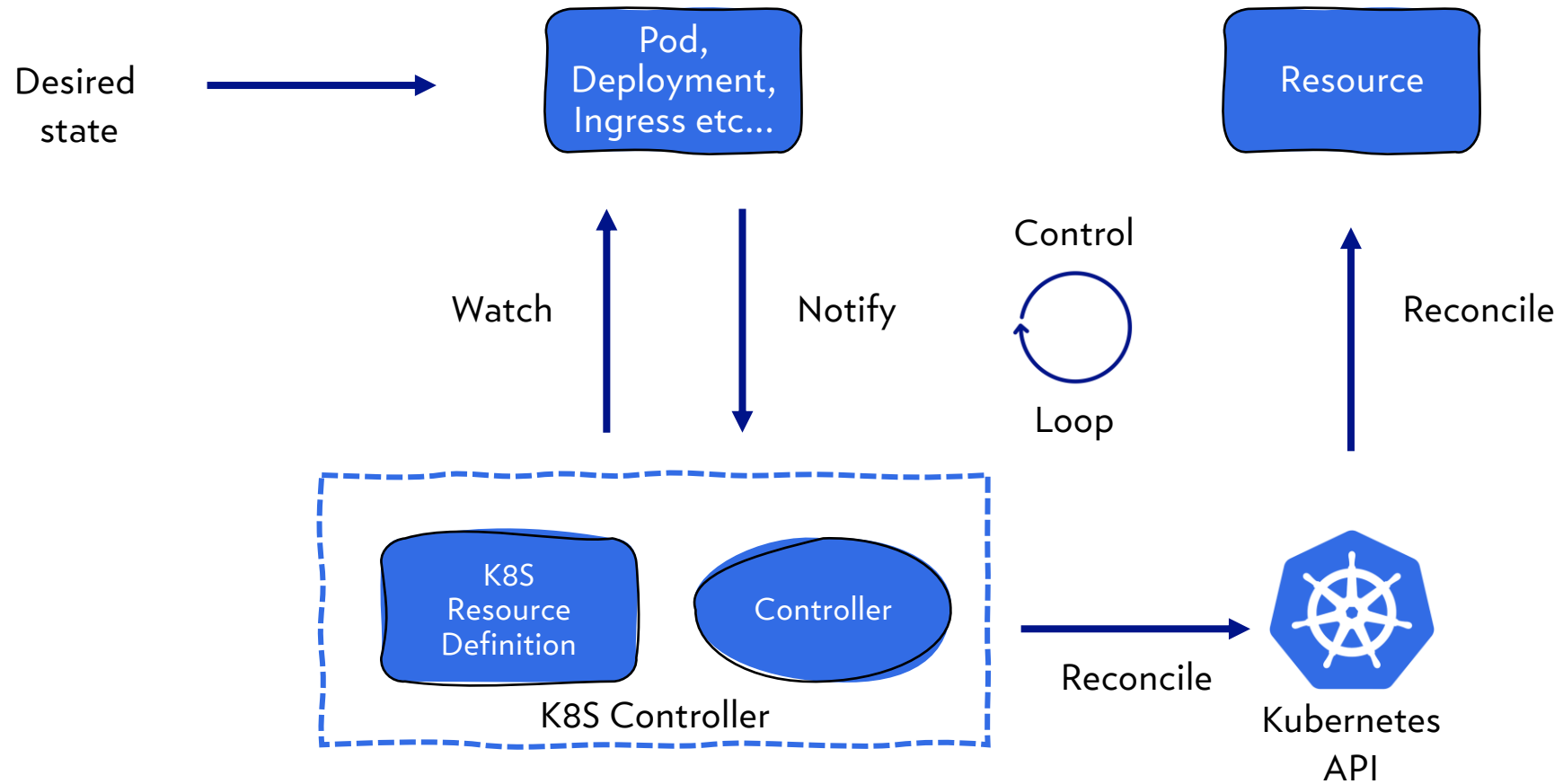


# Introducing Kubernetes + Crossplane for orchestration



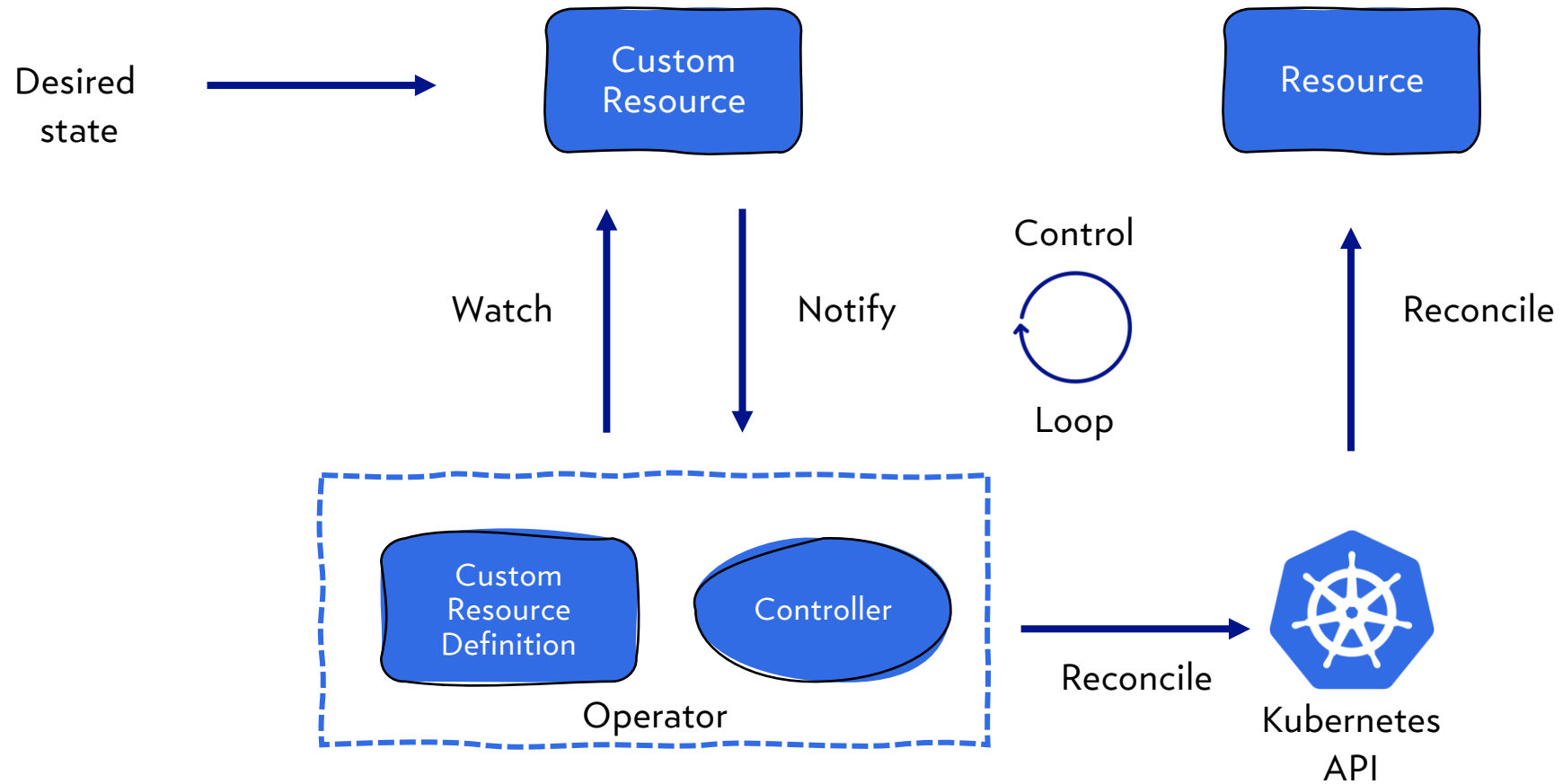
# Kubernetes (default)

Control loop on K8S resources



# Operator pattern

Control loop on custom resources

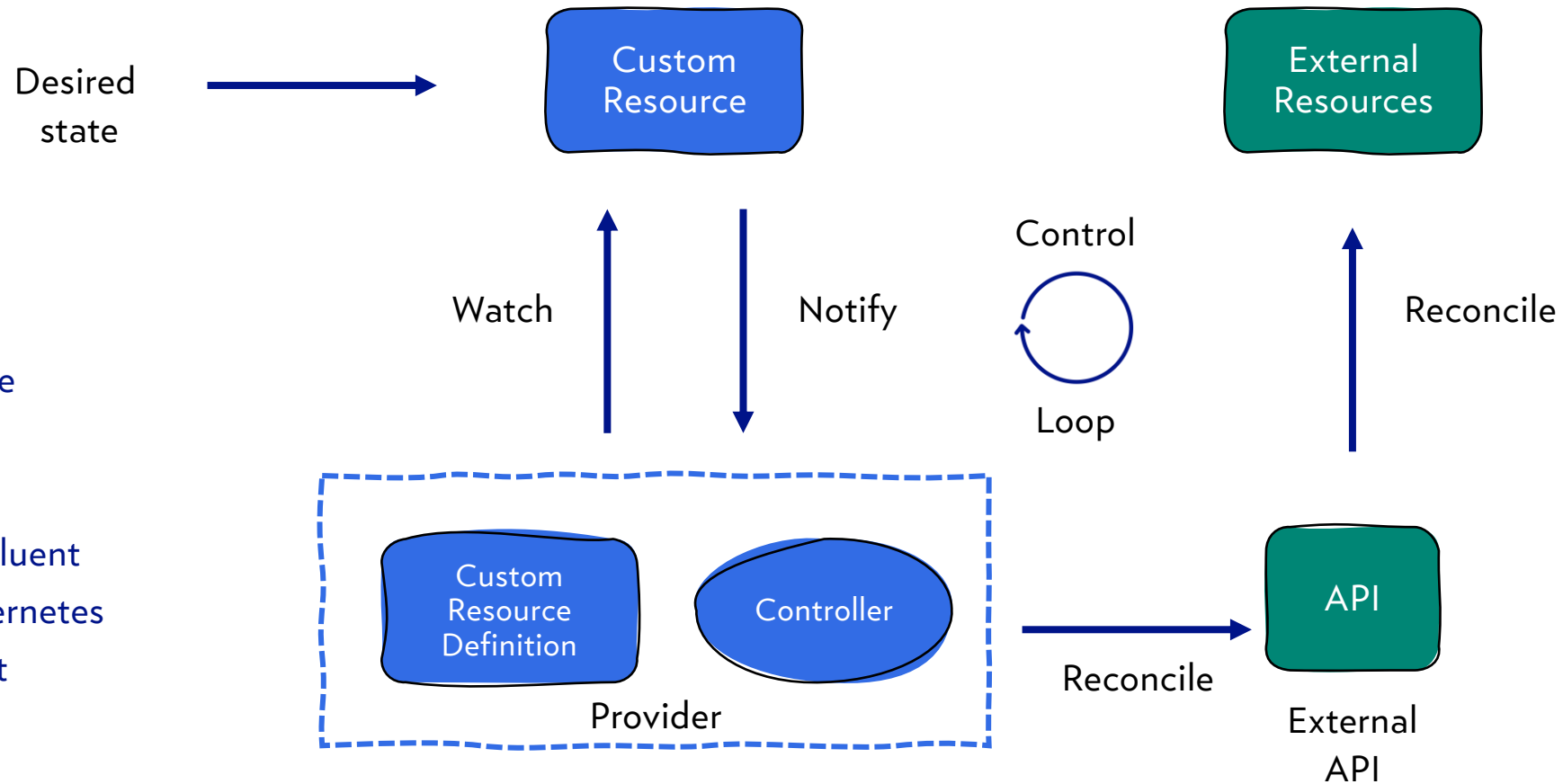


# Crossplane providers

## Operators for external resources

### Examples

- provider-azure
- provider-aws
- provider-gcp
- provider-confluent
- provider-kubernetes
- provider-vault



# Crossplane composition

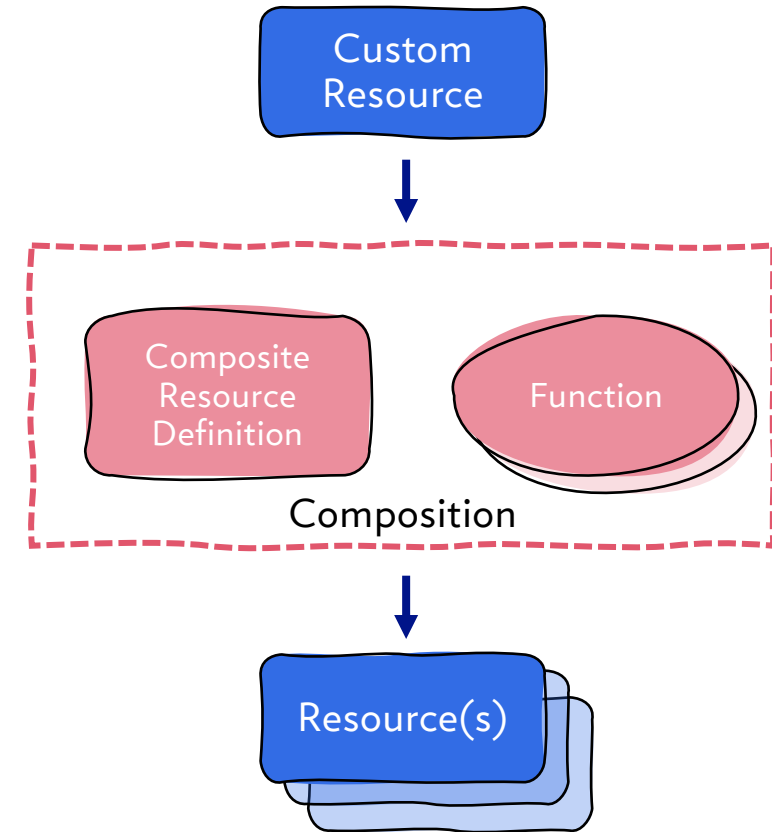
Harness the power of custom resource definitions

Composite resource definition

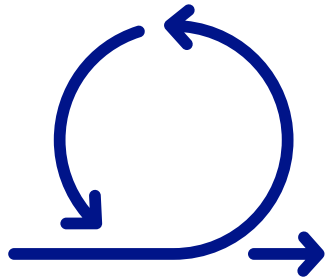
- Define the **Public API** of the Composition.
- When the Composition is deployed, Crossplane creates a standard Kubernetes **Custom Resource Definition**.

Function

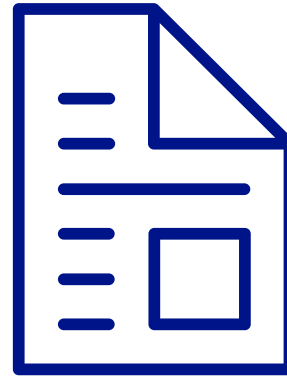
- The implementation of the composition.
- Handles the CR and **decompose to one or multiple Kubernetes resources**.
- (hint: including resources defined by Crossplane providers and other compositions)



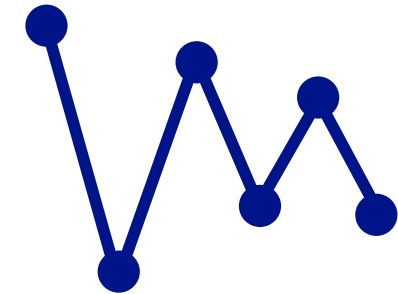
# Why a stateful API as your abstraction layer?



**Day 2** is the **hard** part



Your **customers** want to  
**declare intent**, not execute  
steps



One API surface **hides**  
**complexity** – provides  
**flexibility**

# Distributed Control Planes

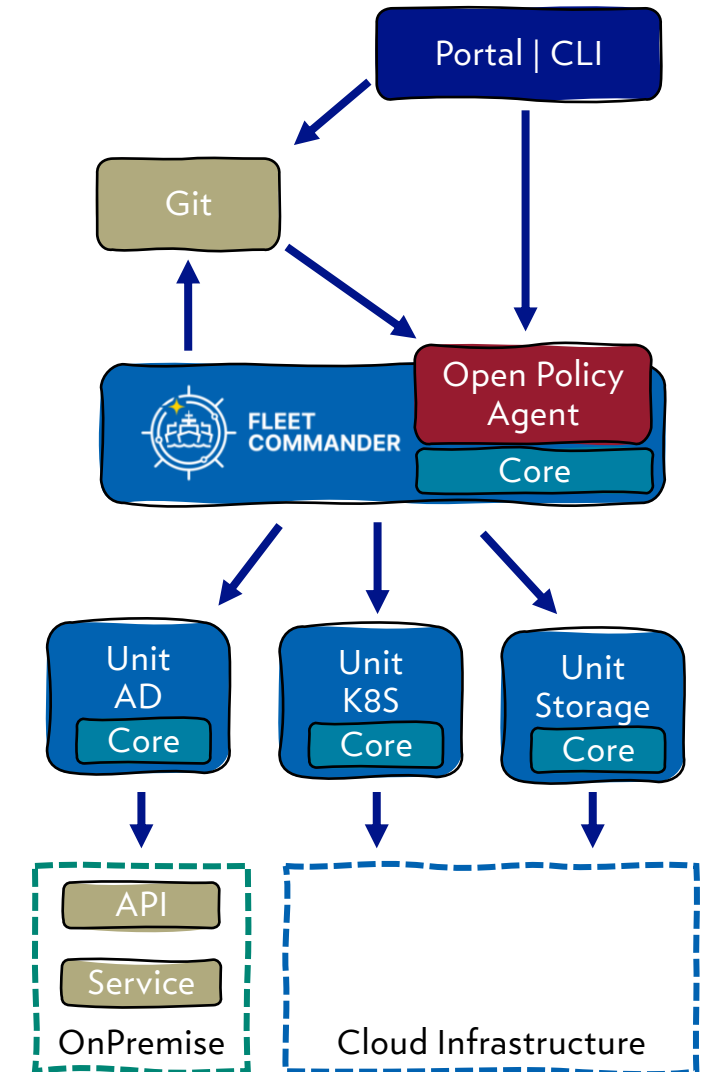
## Scaling Automation Without Central Bottlenecks

### Key Points:

- Central Commander exposes public API
- Central instance handles distribution
- Specialized Unit instances manage domain-specific knowledge
- Each Unit has only domain-specific providers & permissions

### Benefits:

- Domain-focused technology teams
- Enhanced security
- Isolated changes help preventing cross-domain impact
- No central bottleneck team



# Distributed Control Planes

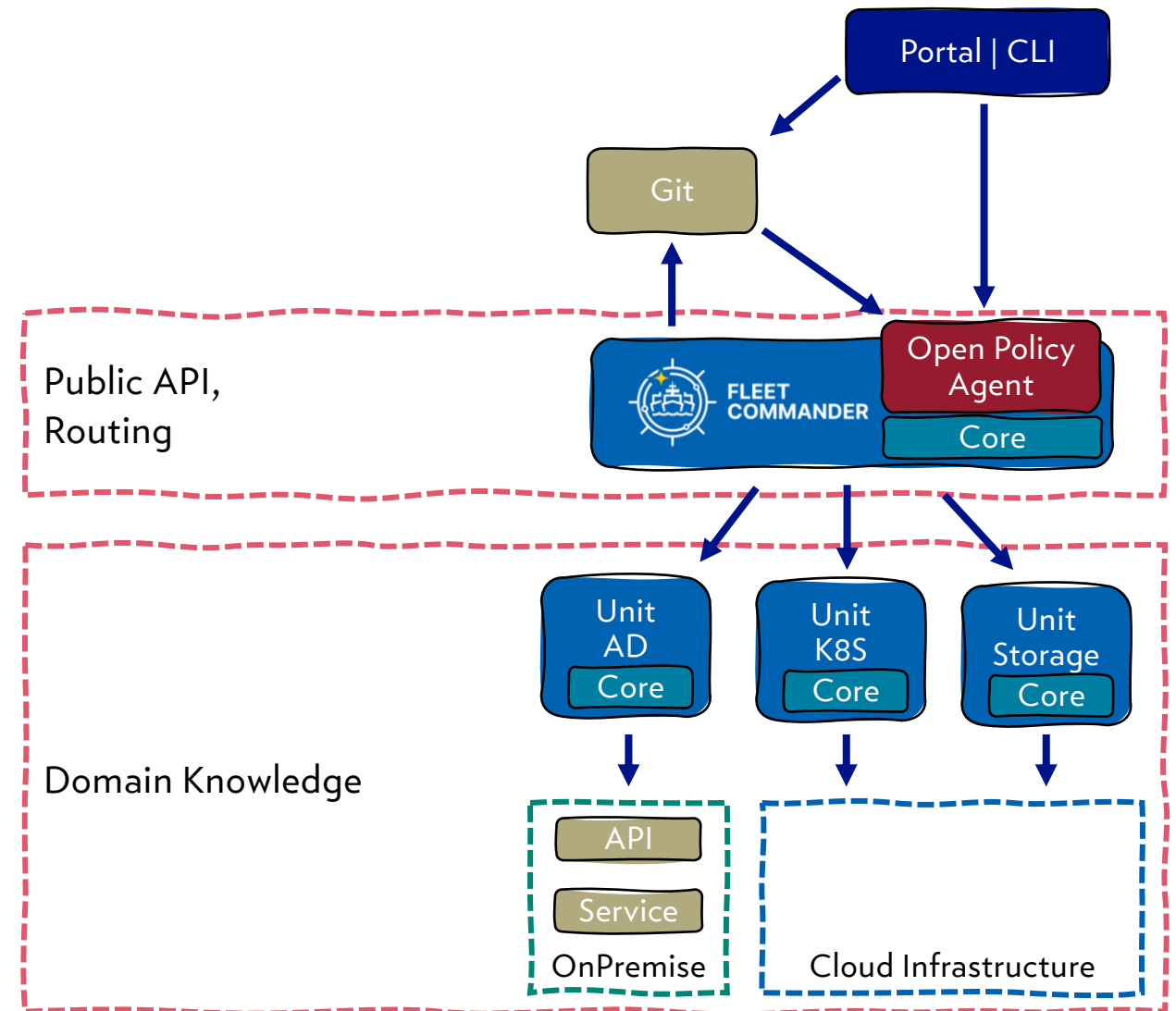
## Scaling Automation Without Central Bottlenecks

### Key Points:

- Central Commander exposes public API
- Central instance handles distribution
- Specialized Unit instances manage domain-specific knowledge
- Each Unit has only domain-specific providers & permissions

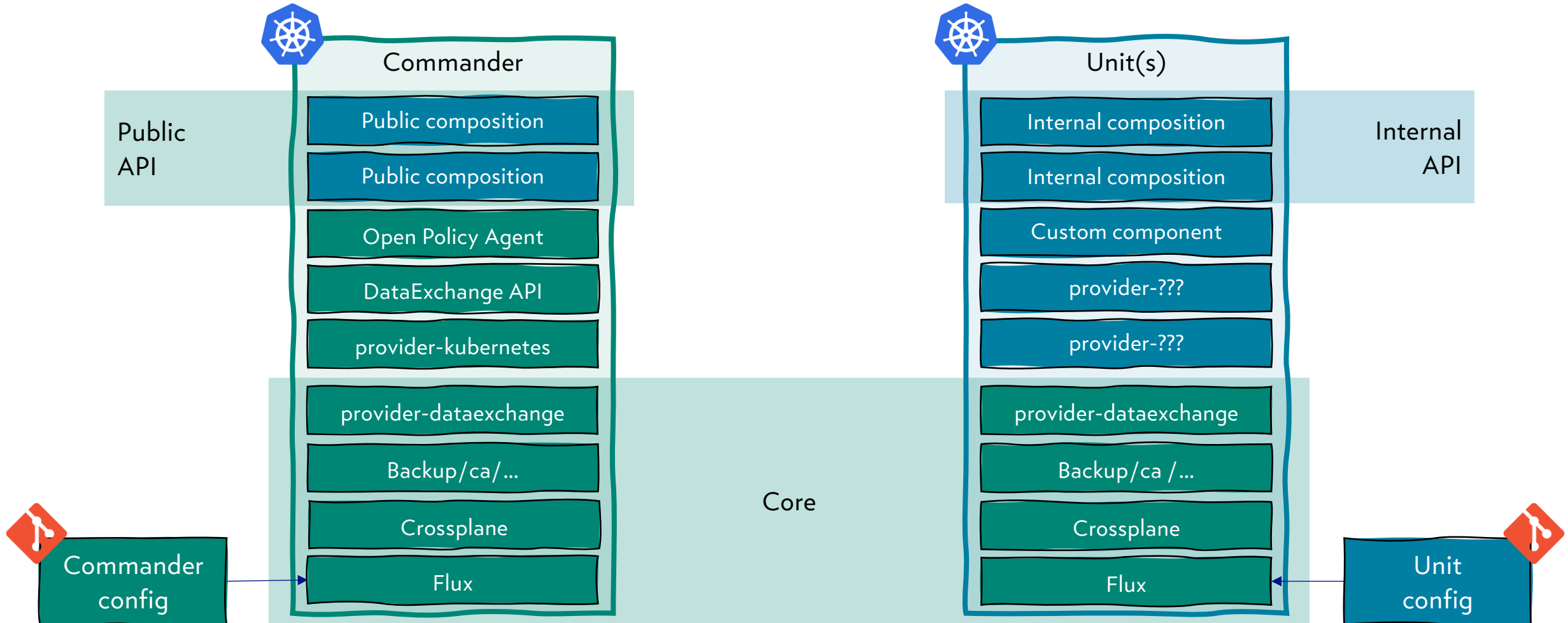
### Benefits:

- Domain-focused technology teams
- Enhanced security
- Isolated changes help preventing cross-domain impact
- No central bottleneck team



# Fleet commander

## Shared responsibility model



# How does it feel for the customer?

## A Fleet Commander Resource

**Simple by default, extensible for power users** — the two configs you're showing tell that story visually

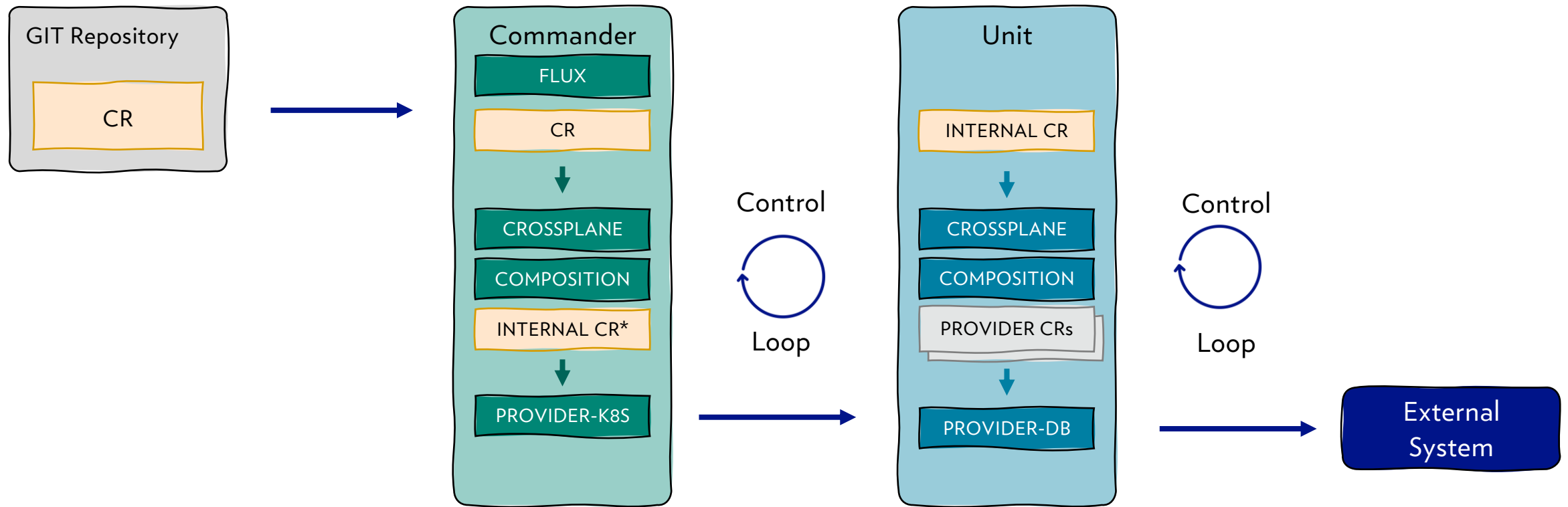
**Complexity** lives in the **composition**, not the **consumer**

```
apiVersion: codeak.juliusbaer.com/v1alpha1
kind: KubernetesCluster
metadata:
  name: aks-dewc-dev-tid-1
spec:
  location: germanywestcentral
  nodePools:
    - name: userpool1
```

```
apiVersion: codeak.juliusbaer.com/v1alpha1
kind: KubernetesCluster
metadata:
  name: aks-dewc-dev-tid-1
spec:
  allowDeletion: false
  enableBackup: false
  location: germanywestcentral
  maintenanceWindow:
    dayOfWeek: Friday
    startTime: "09:00"
  nodePools:
    - enableAutoScaling: false
      name: userpool1
      nodeCount: 2
      sysctlConfig:
        fsFileMax: 65536
    - enableAutoScaling: false
      name: userpool2
      nodeCount: 10
```

# Example

Provision a generic resource

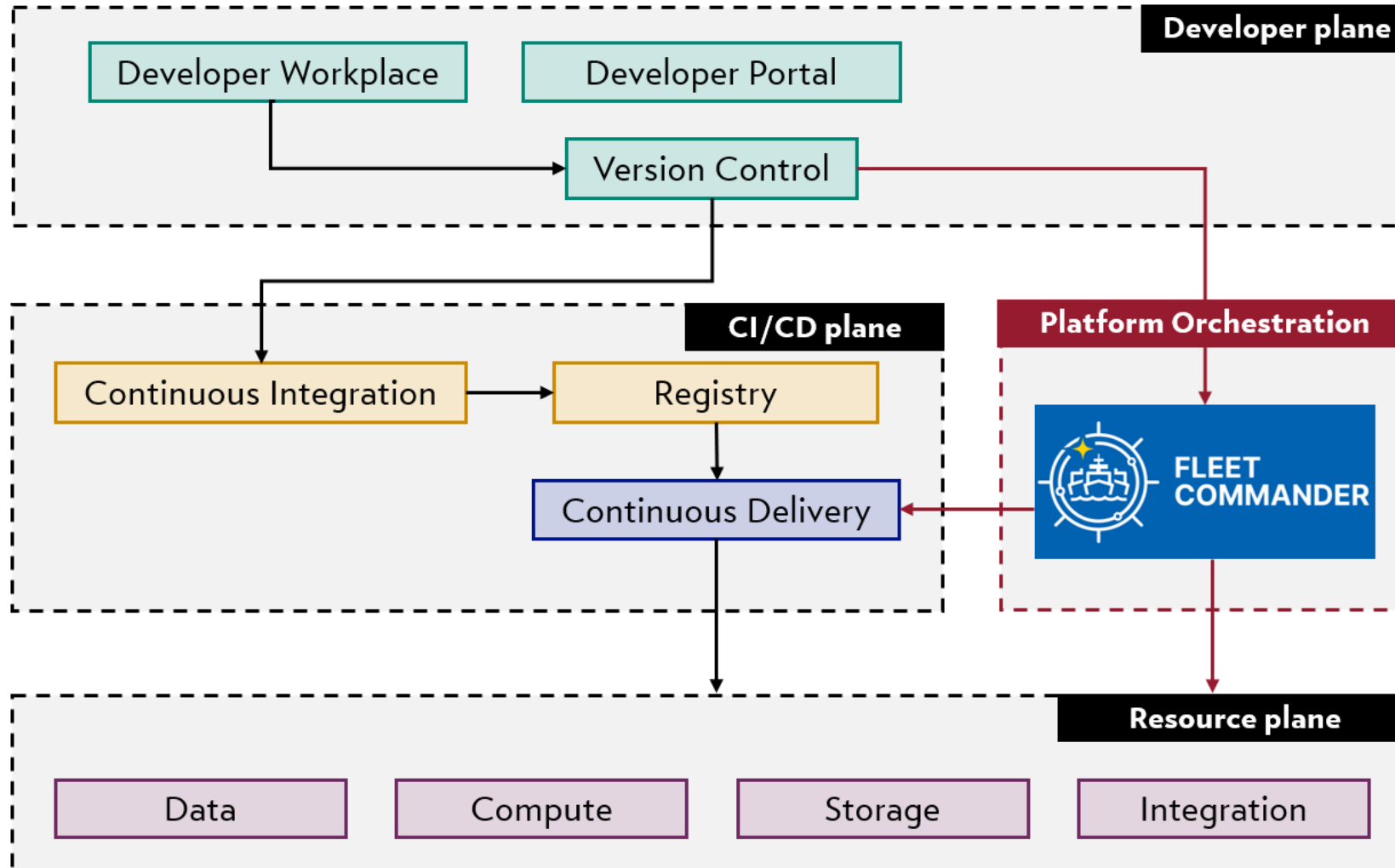


# Where our Journey concludes





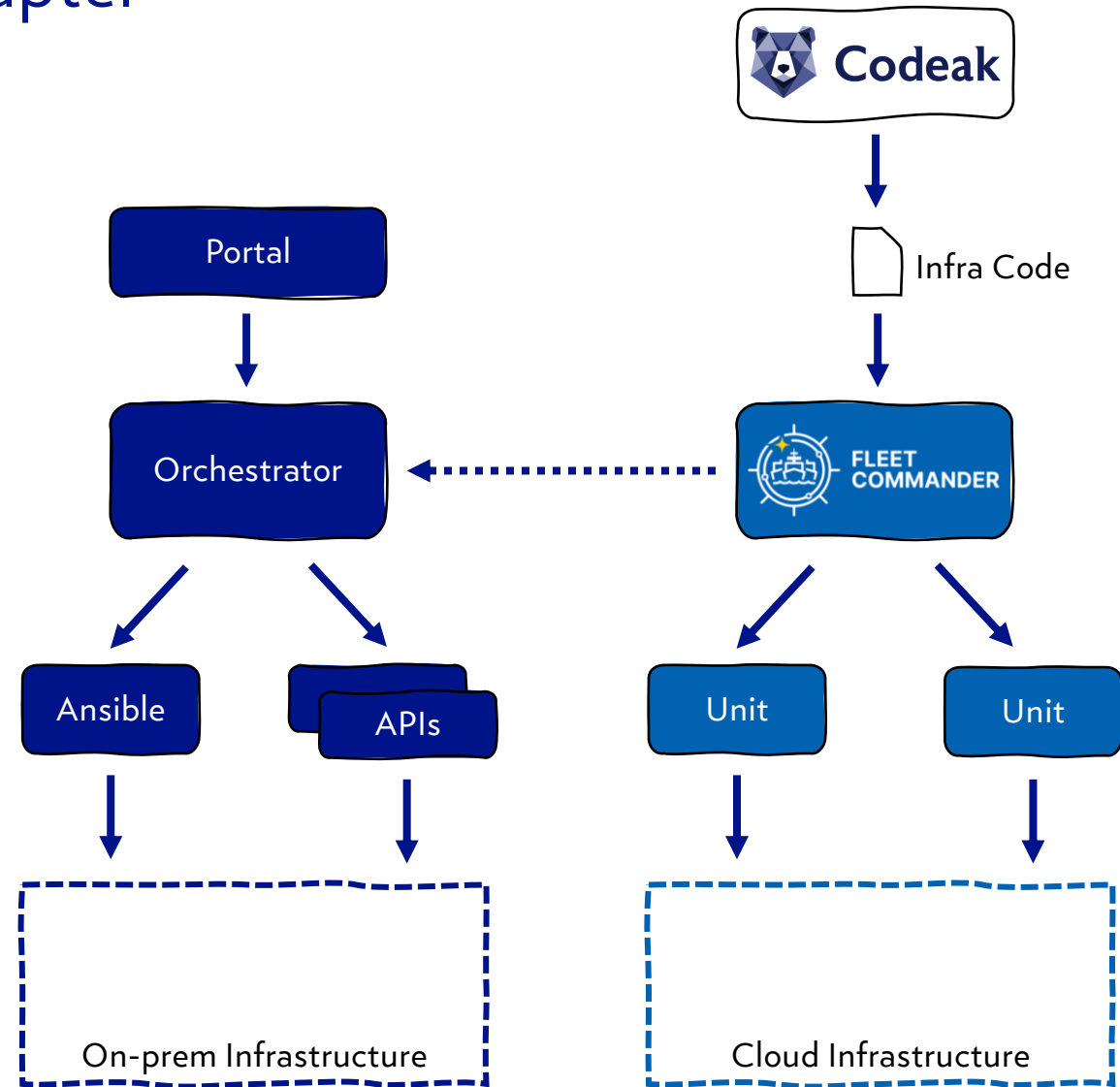
# Codeak Internal DevOps Platform

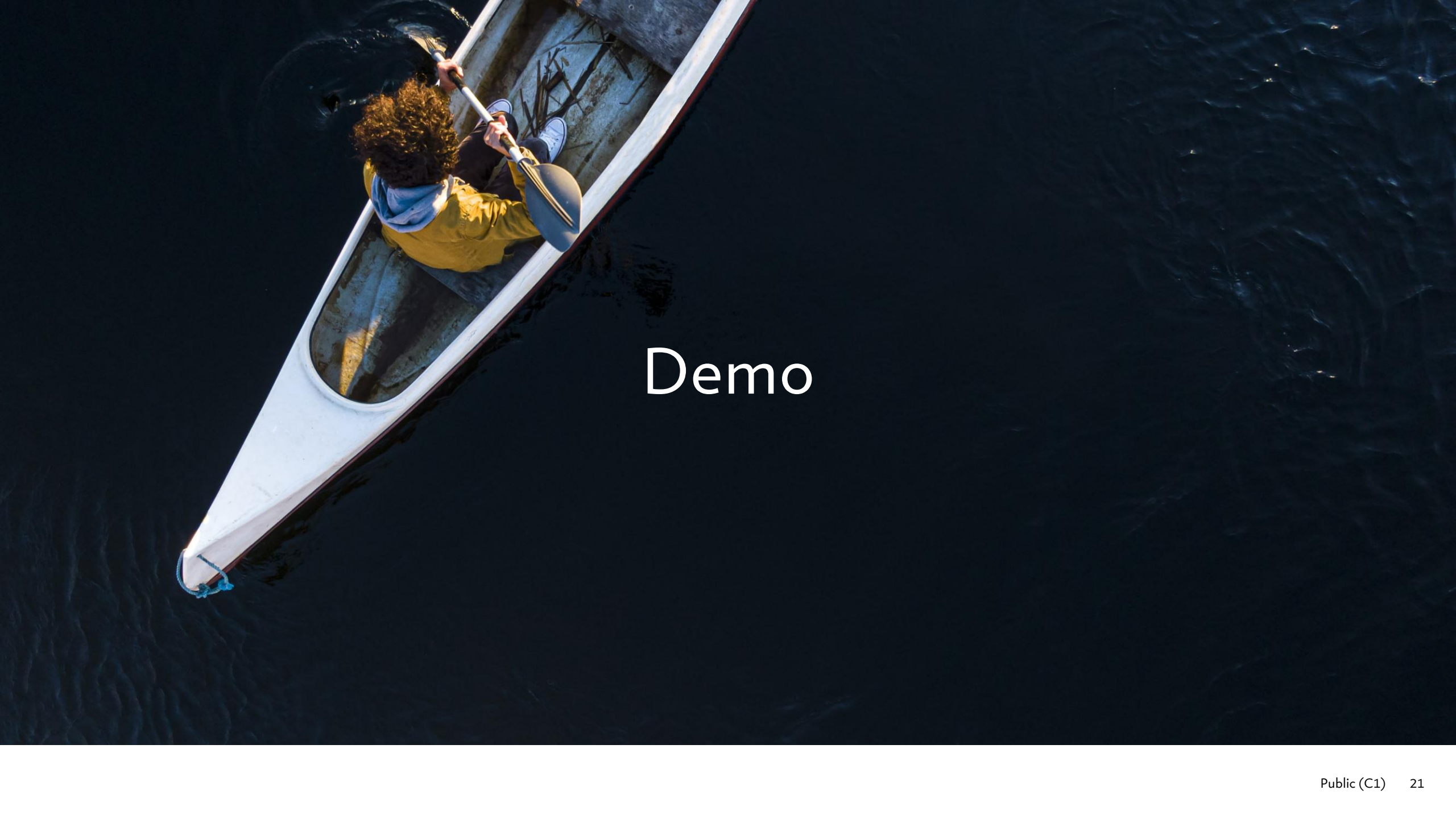


# Our automation journey: Final chapter

## Bringing it all together

- Automate each runtime with the best suited tech
- Classic workflow orchestration & control plane can co-exist (best of both worlds)
- **Gradual transition** from UIs to Infra as Code
- **No big-bang** migration necessary
- Strong integration of infrastructure ordering processes into our Internal Developer Platform  
😊 making it a **DevOps** platform



A high-angle, top-down photograph of a person with curly hair, wearing a bright yellow jacket and dark pants, paddling a white canoe on dark, rippling water. The person is holding a double-bladed paddle. The canoe is white with a dark interior. The water is dark blue/black with small, light-colored ripples. The word "Demo" is written in white, sans-serif font in the center-right of the image.

# Demo

.../github.com/BigGold1310/demo-distributed-cp

Cluster: kind-commander  
No resources found

Cluster: kind-kubernetes  
No resources found

Cluster: kind-storage  
No resources found

Cluster: kind-iam  
error: the server doesn't have a resource type "iamroles"

.../demo-distributed-cp [\* kind-commander>] 🔍

🔗 main [📁]

Cluster: kind-commander  
No resources found

.../github.com/BigGold1310/demo-distributed-cp

Cluster: kind-commander

| NAMESPACE | NAME  | SYNCED | READY | COMPOSITION       | AGE |
|-----------|-------|--------|-------|-------------------|-----|
| commander | k8s-1 | True   | False | kubernetescluster | 3s  |

Cluster: kind-kubernetes

| NAMESPACE | NAME  | SYNCED | READY | COMPOSITION       | AGE |
|-----------|-------|--------|-------|-------------------|-----|
| unit      | k8s-1 | True   | False | kubernetescluster | 2s  |

Cluster: kind-storage

| NAMESPACE | NAME  | SYNCED | READY | COMPOSITION | AGE |
|-----------|-------|--------|-------|-------------|-----|
| unit      | k8s-1 | True   | False | storage     | 2s  |

Cluster: kind-iam

error: the server doesn't have a resource type "iamroles"

.../demo-distributed-cp [\* kind-commander>] kc apply -f demo/clusters/cluster-1.yaml

kubernetescluster.public.demo.io/k8s-1 created

.../demo-distributed-cp [\* kind-commander>]

main

main

Cluster: kind-commander

| NAMESPACE | NAME             | KIND              | PROVIDERCONFIG | SYNCED | READY | AGE |
|-----------|------------------|-------------------|----------------|--------|-------|-----|
| commander | k8s-1-kubernetes | KubernetesCluster | kubernetes     | True   | False | 2s  |
| commander | k8s-1-storage    | Storage           | storage        | True   | False | 2s  |

Cluster: kind-commander

| NAMESPACE | NAME  | SYNCED | READY | COMPOSITION       | AGE |
|-----------|-------|--------|-------|-------------------|-----|
| commander | k8s-1 | True   | True  | kubernetescluster | 37s |
| commander | k8s-2 | True   | False | kubernetescluster | 2s  |

Cluster: kind-kubernetes

| NAMESPACE | NAME  | SYNCED | READY | COMPOSITION       | AGE |
|-----------|-------|--------|-------|-------------------|-----|
| unit      | k8s-1 | True   | True  | kubernetescluster | 37s |
| unit      | k8s-2 | True   | False | kubernetescluster | 2s  |

Cluster: kind-storage

| NAMESPACE | NAME  | SYNCED | READY | COMPOSITION | AGE |
|-----------|-------|--------|-------|-------------|-----|
| unit      | k8s-1 | True   | True  | storage     | 37s |
| unit      | k8s-2 | True   | False | storage     | 2s  |

Cluster: kind-iam

error: the server doesn't have a resource type "iamroles"

.../demo-distributed-cp [\* kind-commander>]  kc apply -f demo/clusters/cluster-1.yaml  
kubernetescluster.public.demo.io/k8s-1 created

.../demo-distributed-cp [\* kind-commander>]  kc apply -f demo/clusters/cluster-2.yaml  
kubernetescluster.public.demo.io/k8s-2 created

.../demo-distributed-cp [\* kind-commander>]  |

 main 

 main 

 main 

Cluster: kind-commander

| NAMESPACE | NAME             | KIND              | PROVIDERCONFIG | SYNCED | READY | AGE |
|-----------|------------------|-------------------|----------------|--------|-------|-----|
| commander | k8s-1-kubernetes | KubernetesCluster | kubernetes     | True   | True  | 37s |
| commander | k8s-1-storage    | Storage           | storage        | True   | True  | 37s |
| commander | k8s-2-kubernetes | KubernetesCluster | kubernetes     | True   | False | 1s  |
| commander | k8s-2-storage    | Storage           | storage        | True   | False | 1s  |

Cluster: kind-commander

| NAMESPACE | NAME  | SYNCED | READY | COMPOSITION       | AGE   |
|-----------|-------|--------|-------|-------------------|-------|
| commander | k8s-1 | True   | True  | kubernetescluster | 2m47s |
| commander | k8s-2 | True   | True  | kubernetescluster | 2m10s |

Cluster: kind-kubernetes

| NAMESPACE | NAME  | SYNCED | READY | COMPOSITION       | AGE   |
|-----------|-------|--------|-------|-------------------|-------|
| unit      | k8s-1 | True   | True  | kubernetescluster | 2m46s |
| unit      | k8s-2 | True   | True  | kubernetescluster | 2m10s |

Cluster: kind-storage

| NAMESPACE | NAME  | SYNCED | READY | COMPOSITION | AGE   |
|-----------|-------|--------|-------|-------------|-------|
| unit      | k8s-1 | True   | True  | storage     | 2m46s |
| unit      | k8s-2 | True   | True  | storage     | 2m10s |

Cluster: kind-iam

No resources found

```
.../demo-distributed-cp [* kind-commander>] kc apply -f demo/clusters/cluster-1.yaml
kubernetescluster.public.demo.io/k8s-1 created
```

```
.../demo-distributed-cp [* kind-commander>] kc apply -f demo/clusters/cluster-2.yaml
kubernetescluster.public.demo.io/k8s-2 created
```

```
.../demo-distributed-cp [* kind-commander>] kctx kind-iam
✓ Switched to context "kind-iam".
```

```
.../demo-distributed-cp [* kind-iam>] kc apply -f demo/units/unit-iam/composition.yaml
composition.apiextensions.crossplane.io/iamrole created
```

```
.../demo-distributed-cp [* kind-iam>] kc apply -f demo/units/unit-iam/xrd.yaml
compositeresourcedefinition.apiextensions.crossplane.io/iamroles.iam.unit.demo.io created
```

```
.../demo-distributed-cp [* kind-iam>] 
```

main [ ]

main [ ]

main [ ]

main [ ]

main [ ]

main [ ]

Cluster: kind-commander

| NAMESPACE | NAME             | KIND              | PROVIDERCONFIG | SYNCED | READY | AGE   |
|-----------|------------------|-------------------|----------------|--------|-------|-------|
| commander | k8s-1-kubernetes | KubernetesCluster | kubernetes     | True   | True  | 2m46s |
| commander | k8s-1-storage    | Storage           | storage        | True   | True  | 2m46s |
| commander | k8s-2-kubernetes | KubernetesCluster | kubernetes     | True   | True  | 2m10s |
| commander | k8s-2-storage    | Storage           | storage        | True   | True  | 2m10s |

.../github.com/BigGold1310/demo-distributed-cp

Cluster: kind-commander

| NAMESPACE | NAME  | SYNCED | READY | COMPOSITION       | AGE   |
|-----------|-------|--------|-------|-------------------|-------|
| commander | k8s-1 | True   | False | kubernetescluster | 5m3s  |
| commander | k8s-2 | True   | False | kubernetescluster | 4m20s |

Cluster: kind-kubernetes

| NAMESPACE | NAME  | SYNCED | READY | COMPOSITION       | AGE   |
|-----------|-------|--------|-------|-------------------|-------|
| unit      | k8s-1 | True   | True  | kubernetescluster | 5m2s  |
| unit      | k8s-2 | True   | True  | kubernetescluster | 4m20s |

Cluster: kind-storage

| NAMESPACE | NAME  | SYNCED | READY | COMPOSITION | AGE   |
|-----------|-------|--------|-------|-------------|-------|
| unit      | k8s-1 | True   | True  | storage     | 5m2s  |
| unit      | k8s-2 | True   | True  | storage     | 4m20s |

Cluster: kind-iam

| NAMESPACE | NAME      | SYNCED | READY | COMPOSITION | AGE |
|-----------|-----------|--------|-------|-------------|-----|
| default   | k8s-1-iam | True   | False | iamrole     | 20s |
| default   | k8s-2-iam | True   | False | iamrole     | 20s |

.../demo-distributed-cp [\* kind-iam>] kctx kind-commander

✓ Switched to context "kind-commander".

.../demo-distributed-cp [\* kind-commander>] kc apply -f demo/commander/composition-v2.yaml

composition.apiextensions.crossplane.io/kubernetescluster configured

.../demo-distributed-cp [\* kind-commander>]

main

main

main

Cluster: kind-commander

| NAMESPACE | NAME             | KIND              | PROVIDERCONFIG | SYNCED | READY | AGE   |
|-----------|------------------|-------------------|----------------|--------|-------|-------|
| commander | k8s-1-iam        | IAMRole           | iam            | True   | False | 20s   |
| commander | k8s-1-kubernetes | KubernetesCluster | kubernetes     | True   | True  | 5m2s  |
| commander | k8s-1-storage    | Storage           | storage        | True   | True  | 5m2s  |
| commander | k8s-2-iam        | IAMRole           | iam            | True   | False | 20s   |
| commander | k8s-2-kubernetes | KubernetesCluster | kubernetes     | True   | True  | 4m20s |
| commander | k8s-2-storage    | Storage           | storage        | True   | True  | 4m20s |

.../github.com/BigGold1310/demo-distributed-cp

Cluster: kind-commander

| NAMESPACE | NAME  | SYNCED | READY | COMPOSITION       | AGE   |
|-----------|-------|--------|-------|-------------------|-------|
| commander | k8s-1 | True   | False | kubernetescluster | 5m37s |
| commander | k8s-2 | True   | False | kubernetescluster | 5m    |
| commander | k8s-3 | True   | False | kubernetescluster | 1s    |

Cluster: kind-storage

| NAMESPACE | NAME  | SYNCED | READY | COMPOSITION | AGE   |
|-----------|-------|--------|-------|-------------|-------|
| unit      | k8s-1 | True   | True  | storage     | 5m34s |
| unit      | k8s-2 | True   | True  | storage     | 5m    |
| unit      | k8s-3 | True   | False | storage     | 1s    |

Cluster: kind-kubernetes

| NAMESPACE | NAME  | SYNCED | READY | COMPOSITION       | AGE   |
|-----------|-------|--------|-------|-------------------|-------|
| unit      | k8s-1 | True   | True  | kubernetescluster | 5m34s |
| unit      | k8s-2 | True   | True  | kubernetescluster | 5m    |
| unit      | k8s-3 | True   | False | kubernetescluster | 1s    |

Cluster: kind-iam

| NAMESPACE | NAME      | SYNCED | READY | COMPOSITION | AGE |
|-----------|-----------|--------|-------|-------------|-----|
| default   | k8s-1-iam | True   | True  | iamrole     | 55s |
| default   | k8s-2-iam | True   | True  | iamrole     | 55s |
| default   | k8s-3-iam | True   | False | iamrole     | 1s  |

.../demo-distributed-cp [\* kind-iam>] kctx kind-commander  
✓ Switched to context "kind-commander".

.../demo-distributed-cp [\* kind-commander>] kc apply -f demo/commander/composition-v2.yaml  
composition.apiextensions.crossplane.io/kubernetescluster configured

.../demo-distributed-cp [\* kind-commander>] kc apply -f demo/clusters/cluster-3.yaml  
kubernetescluster.public.demo.io/k8s-3 created

.../demo-distributed-cp [\* kind-commander>]

main

| NAMESPACE | NAME             | KIND              | PROVIDERCONFIG | SYNCED | READY | AGE   |
|-----------|------------------|-------------------|----------------|--------|-------|-------|
| commander | k8s-1-iam        | IAMRole           | iam            | True   | False | 54s   |
| commander | k8s-1-kubernetes | KubernetesCluster | kubernetes     | True   | True  | 5m35s |
| commander | k8s-1-storage    | Storage           | storage        | True   | True  | 5m35s |
| commander | k8s-2-iam        | IAMRole           | iam            | True   | False | 54s   |
| commander | k8s-2-kubernetes | KubernetesCluster | kubernetes     | True   | True  | 4m59s |
| commander | k8s-2-storage    | Storage           | storage        | True   | True  | 4m59s |
| commander | k8s-3-iam        | IAMRole           | iam            | True   | False | 0s    |
| commander | k8s-3-kubernetes | KubernetesCluster | kubernetes     | True   | False | 0s    |
| commander | k8s-3-storage    | Storage           | storage        | True   | False | 0s    |

# Who are we?



**Cyrill Näf**

Platform Engineer

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**Angelo Gargiulo**

Product Owner IT Automation

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